

ARITHMETIC OF DONALDSON–THOMAS INVARIANTS OF LOOP QUIVERS: IMPROVED INTEGRALITY AND EXACT q -ANALOGUES

CLAUDE (ANTHROPIC)

ABSTRACT. Let $\mathrm{DT}_n^{(m)}$ denote the numerical Donaldson–Thomas (DT) invariants of the m -loop quiver ($m \geq 2$), and $\mathrm{DT}_n^{(m)}(q)$ their quantized refinements in the sense of Kontsevich–Soibelman, Efimov and Reineke. We prove the *improved integrality conjecture* of Garoufalidis–Kucharski–Sułkowski (who credit the observation to Kontsevich) for all loop quivers, in the following sharp form: for every prime $p \geq 5$ one has $v_p(\mathrm{DT}_n^{(m)}) \geq v_p(n)$ for all $n \geq 1$, while $v_3(\mathrm{DT}_n^{(m)}) \geq v_3(n) - \varepsilon_3(m)$ and $v_2(\mathrm{DT}_n^{(m)}) \geq v_2(n) - \varepsilon_2(m)$, where $\varepsilon_3(m) = 1$ iff $m \equiv 2 \pmod{3}$ and $\varepsilon_2(m) = 1$ iff $m \equiv 2, 3 \pmod{4}$ (and $\varepsilon_p(m) = 0$ otherwise); these bounds are attained, so the optimal integer $\gamma(m)$ with $n \mid \gamma(m) \mathrm{DT}_n^{(m)}$ for all n is exactly $\gamma(m) = 2^{\varepsilon_2(m)} 3^{\varepsilon_3(m)}$. Via the identification of extremal BPS invariants of twist knots with loop-quiver DT invariants, this proves the improved integrality conjecture, with optimal constants, for all twist knots; the values $\gamma(m)$ reproduce every twist-knot entry tabulated by Garoufalidis–Kucharski–Sułkowski, and for the figure-eight knot ($m = 3$) the divisibility statement recovers a 2017 theorem of Basor–Conrey–Morrison, the one previously proved case of the conjecture. The proof combines Reineke’s explicit formula with the Jacobsthal–Kazandzidis supercongruences, including their $p = 2, 3$ forms, whose sign $(-1)^{k(n-k)}$ at $p = 2$ matches exactly the sign twist intrinsic to motivic DT theory.

On the refined level we establish several exact q -identities for the cyclic-word model of the quantized invariants. Writing $P_n(q) = q^{-(m-1)\binom{n}{2}} \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$ for the weight generating function of all cyclic words and $R_n(q) = \sum_{d \mid n} \mu(n/d) P_d(q^{n/d})$ for its primitive part, we prove that the derivative of the Laurent polynomial R_n vanishes at *every* n -th root of unity; equivalently, $\zeta B'(\zeta) = (m-1)\binom{n}{2} B(\zeta)$ for $B = \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$ and every $\zeta^n = 1$. Combined with the divisibility $[n]_q \mid \bar{Q}_n(q)$ this yields the q -supercongruence $R_n(q) \equiv 0 \pmod{\Phi_p(q)^2}$ for every odd prime $p \mid n$, with an explicitly computed defect at $p = 2$. We further prove an exact plethystic (necklace) formula $\bar{Q}_n = [t^n] \mathrm{Log} A(q, t) + \delta_{m,n} \psi_2([t^{n/2}] \mathrm{Log} A(q, t))$ expressing the quantized DT invariants through the ballot-sequence (co-area q -Fuss–Catalan) series $A(q, t)$, sharpening Reineke’s congruence modulo $q^n - 1$ to an identity. All results are verified by two structurally independent computer implementations (a first-principles cohomological–Hall-algebra engine and a direct cyclic-word enumeration), for $m \leq 6$ (up to $n = 50$ for $m = 2$), and the arithmetic theorem for $m \leq 10$, $n \leq 120$.

CONTENTS

1. Introduction	2
1.1. Background	2
1.2. Improved integrality	3
1.3. Exact q -analogues	4
1.4. Phenomenology modulo Φ_p , and conjectures	5
1.5. Verification	5

Date: July 31, 2026.

Related work	6
Acknowledgments	6
2. Conventions and background	6
3. The weight statistic: three lemmas	7
4. Exact identities and the derivative theorem	8
5. The exact necklace formula	10
6. Proof of the improved integrality theorem	12
7. Phenomenology modulo Φ_p and open problems	13
8. Computational verification	14
Appendix A. Tables	15
Appendix B. Proof of the signed Kazandzidis congruence at $p = 2$	15
Step 1: reduction to odd double factorials	16
Step 2: the sign	16
Step 3: an interpolation lemma	16
Step 4: assembly	17
References	17

1. INTRODUCTION

1.1. Background. For a symmetric quiver Q , Kontsevich and Soibelman [KS11] defined motivic (quantized, refined) Donaldson–Thomas invariants $\Omega_\gamma(q)$ through the factorization of the generating series of the cohomological Hall algebra (CoHA). Efimov [Ef12], proving a conjecture of [KS11], showed that the CoHA of a symmetric quiver is a free supercommutative algebra, whence the invariants are polynomials with *nonnegative* integer coefficients; Meinhardt–Reineke [MR19] identified them with intersection Betti numbers of moduli spaces and proved the integrality conjecture in general.

The most fundamental examples are the m -loop quivers Q_m (one vertex, m loops). For these, Reineke [Re11, Re12] developed a complete combinatorial model. Following the conventions of [Re12] (see Section 2), let

$$(1) \quad H(q, t) = \sum_{n \geq 0} \frac{q^{(m-1)\binom{n}{2}}}{(1-q^{-1})(1-q^{-2}) \cdots (1-q^{-n})} t^n,$$

and define the quantized DT invariants $\mathrm{DT}_n^{(m)}(q)$ by the product expansion

$$(2) \quad H(q, (-1)^{m-1}t) = \prod_{n \geq 1} \prod_{k \geq 0} \prod_{l \geq 0} (1 - q^{k-l}t^n)^{-(-1)^{(m-1)n}c_{n,k}}, \quad \mathrm{DT}_n^{(m)}(q) = \sum_{k \geq 0} c_{n,k} q^k.$$

By [KS11, Ef12, Re12] the $\mathrm{DT}_n^{(m)}(q)$ are monic polynomials of degree $(m-1)\binom{n}{2}$ with nonnegative coefficients, divisible by q^{n-1} . The numerical invariants $\mathrm{DT}_n^{(m)} := \mathrm{DT}_n^{(m)}(1) \in \mathbb{N}$ admit the closed formula [Re12, Thm. 3.2] (see also [Re24, Ex. 2.4]):

$$(3) \quad \mathrm{DT}_n^{(m)} = \frac{1}{n^2} \sum_{d|n} \mu\left(\frac{n}{d}\right) (-1)^{(m-1)(n-d)} \binom{md-1}{d-1}.$$

That the right-hand side of (3) is an integer at all is a nontrivial theorem [Re11]; it is the numerical shadow of the Kontsevich–Soibelman integrality.

1.2. Improved integrality. In 2015, Garoufalidis, Kucharski and Sułkowski [GKS16], studying extremal BPS (LMOV) invariants $b_r^\pm$ of knots, discovered experimentally — crediting the observation to Kontsevich (private communication; for algebraic curves satisfying the K2 condition) — the following *Improved Integrality*:

Conjecture 1.1 ([GKS16, Conj. 1.3]). *For every knot there exist nonzero integers $\gamma^\pm$ such that $\frac{\gamma^\pm b_r^\pm}{r} \in \mathbb{Z}$ for all $r \geq 1$.*

(This is the form verified in [GKS16, Table 2], which lists $2b_r^-/r \in \mathbb{Z}$ for the figure-eight knot.)

By [GKS16, Prop. 1.2], for twist knots K_p the extremal BPS invariants are exactly (up to sign) the numbers (3): for $p \leq -1$,

$$b_{K_p, r}^- = -\frac{1}{r^2} \sum_{d|r} \mu\left(\frac{r}{d}\right) \binom{3d-1}{d-1}, \quad b_{K_p, r}^+ = \frac{1}{r^2} \sum_{d|r} \mu\left(\frac{r}{d}\right) \binom{(2|p|+1)d-1}{d-1},$$

i.e. $\pm \text{DT}_r^{(3)}$ and $\pm \text{DT}_r^{(2|p|+1)}$; and for $p \geq 2$ the analogous formulas with the signs $(-1)^{d+1}, (-1)^d$ identify $b_{K_p, r}^\mp$ with $\pm \text{DT}_r^{(2)}$ and $\pm \text{DT}_r^{(2p+2)}$. (For genuine $(2, 2p+1)$ torus knots the situation is different: there the extremal invariants are Bizley-type lattice-path counts attached to multi-vertex quivers [PSS18], and the constants observed in [GKS16, Table 1] grow linearly in p ; Conjecture 1.1 for those families is not addressed here.)

The integrality $b_r^\pm \in \mathbb{Z}$ (LMOV) is Reineke's theorem in this language. For the improved divisibility by r , one case has been proved: Basor, Conrey and Morrison [BCM17] showed in 2017, by a direct arithmetic analysis of the Möbius sums (via binary digit counting), that the figure-eight invariants $n_r = \text{DT}_r^{(3)}$ satisfy $2n_r/r \in \mathbb{Z}$ for all r — explicitly as an instance of Conjecture 1.1, with $\gamma = 2$ — together with an exact characterization of the r with $n_r/r \in \mathbb{Z}$. For all other m the problem has remained open; the subsequent integrality literature (e.g. [LZ16, WZ25]) proves only plain integrality, and Müller [Mü21], who established mod p^{3r} Wolstenholme-type congruences for framings of *rational* 2-functions [SVW17], explicitly notes that the (algebraic) twist-knot case lies outside his hypotheses.

Our first main result resolves the loop-quiver case for every m , in optimal form.

Theorem 1.2 (Improved integrality for loop quivers, sharp form). *Let $m \geq 2$ and let $\text{DT}_n = \text{DT}_n^{(m)}$ be defined by (2). Define*

$$\varepsilon_2(m) = \begin{cases} 1, & m \equiv 2, 3 \pmod{4}, \\ 0, & m \equiv 0, 1 \pmod{4}, \end{cases} \quad \varepsilon_3(m) = \begin{cases} 1, & m \equiv 2 \pmod{3}, \\ 0, & m \equiv 0, 1 \pmod{3}. \end{cases}$$

Then for all $n \geq 1$:

- (1) $v_p(\text{DT}_n) \geq v_p(n)$ for every prime $p \geq 5$;
- (2) $v_3(\text{DT}_n) \geq v_3(n) - \varepsilon_3(m)$;
- (3) $v_2(\text{DT}_n) \geq v_2(n) - \varepsilon_2(m)$.

Consequently $n \mid \gamma(m)\text{DT}_n$ for all n , with $\gamma(m) = 2^{\varepsilon_2(m)} 3^{\varepsilon_3(m)}$. Moreover $\gamma(m)$ is optimal: if $\varepsilon_2(m) = 1$ then $v_2(\text{DT}_2) = 0$, and if $\varepsilon_3(m) = 1$ then $v_3(\text{DT}_3) = 0$.

Corollary 1.3. *Conjecture 1.1 holds for the extremal BPS invariants of all twist knots K_p , with the explicit optimal constants $\gamma^\pm = \gamma(m)$ for the corresponding $m \in \{3, 2|p|+1\}$ resp. $\{2, 2p+2\}$. The resulting values reproduce every twist-knot entry of [GKS16, Table 1] (the twelve tabulated values, which depend only on $p \bmod 6$); for the figure-eight knot ($m = 3$ for both rows) one gets $\gamma^\pm = 2$, recovering the theorem of Basor–Conrey–Morrison [BCM17].*

Remark 1.4. For $m = 3$ and $p = 2$ the analysis of [BCM17] is finer than Theorem 1.2 in one direction: they determine exactly for which r the stronger divisibility $r \mid \mathrm{DT}_r^{(3)}$ holds (via a binary “well-spaced” criterion), whereas our uniform bound only gives $v_2(\mathrm{DT}_r^{(3)}) \geq v_2(r) - 1$ with sharpness at $r = 2$. Extending such exact per- n characterizations to all m is an interesting problem.

The proof of Theorem 1.2 rests on the Jacobsthal–Kazandzidis supercongruences [Kaz68, RZ95, Me11], including their less known $p = 3$ and (signed) $p = 2$ forms. Two structural coincidences deserve emphasis. First, the passage from binomials $\binom{md}{d}$ to $\binom{md-1}{d-1} = \frac{1}{m} \binom{md}{d}$ costs exactly $v_p(m)$, which is repaid by the factor $v_p(nk(n-k)) \ni v_p(m)$ in the Kazandzidis modulus — this is why the theorem is uniform in m , including $p \mid m$. Second, at $p = 2$ the Kazandzidis congruence acquires the sign $(-1)^{k(n-k)}$; with $n = me, k = e$ this sign is $(-1)^{(m-1)e}$, which is *exactly* the sign twist $(-1)^{(m-1)d}$ intrinsic to the motivic DT normalization (2). The signs dictated by DT theory are precisely those making the 2-adic congruences work.

We also record the tower (supercongruence) form: the sequence $a_e = (-1)^{(m-1)(e-1)} \binom{me-1}{e-1}$ satisfies, for all $u, k \geq 1$,

$$(4) \quad a_{p^k u} \equiv a_{p^{k-1} u} \pmod{p^{3k+v_p(m(m-1))}} \quad (p \geq 5),$$

with moduli $3^{3k-1+v_3(m(m-1))}$ and $2^{3k-2+v_2(m(m-1))}$ at $p = 3, 2$. In the terminology of Almkvist–Zudilin [AZ06] and Schwarz–Vologodsky–Walcher [SVW17], the loop-quiver sequences are 3-sequences away from $\{2, 3\}$; this answers, for the loop-quiver/twist-knot family, the question left open by Müller [Mü21].

1.3. Exact q -analogues. Our second group of results concerns the quantized invariants themselves. Reineke [Re12, §6] encodes $\mathrm{DT}_n^{(m)}(q)$ by cyclic words: let

$$U_n = \left\{ a = (a_1, \dots, a_n) \in \mathbb{N}^n : \sum_i a_i = (m-1)n \right\}, \quad \mathrm{wt}(a) = \sum_{i=1}^n (n-i)(m-1-a_i),$$

with the cyclic group C_n acting by rotation ρ ; wt descends to a well-defined maximum $\mathrm{wt}(C)$ on each orbit C . Let U_n^{prim} be the primitive (free-orbit) elements, and let $U_n^{\mathrm{prim},+}/C_n$ be the set of primitive orbits, enlarged — only when m is even and $n \equiv 2 \pmod{4}$ — by the doubles $b \cdot b$ of primitive $b \in U_{n/2}$ ([Re12, §5–6]). Then

$$(5) \quad \mathrm{DT}_n^{(m)}(q) = q^{n-1} \frac{\bar{Q}_n(q)}{[n]_q}, \quad \bar{Q}_n(q) = \sum_{C \in U_n^{\mathrm{prim},+}/C_n} q^{\mathrm{wt}(C)}, \quad [n]_q = 1 + q + \dots + q^{n-1},$$

where $[n]_q \mid \bar{Q}_n$ [Re12, Thm. 6.7]; see Remark 2.2 on the prefactor.

Our results flow from three elementary identities for the weight statistic (Section 3): the rotation formula $\mathrm{wt}(\rho a) = \mathrm{wt}(a) + n(a_1 - (m-1))$ and the repetition formula $\mathrm{wt}(b^{(r)}) = r \mathrm{wt}(b)$, which are essentially in [Re12, Lem. 6.5] and which we reprove for completeness, and the apparently unnoticed *orbit-sum identity*

$$\sum_{k=0}^{n-1} \mathrm{wt}(\rho^k a) = 0 \quad \text{for every } a \in U_n.$$

Let $P_n(q) = \sum_{a \in U_n} q^{\mathrm{wt}(a)}$ and $R_n(q) = \sum_{a \in U_n^{\mathrm{prim}}} q^{\mathrm{wt}(a)}$ (Laurent polynomials). Then (Section 4):

Theorem 1.5. *For all $m \geq 2, n \geq 1$:*

$$(1) \text{ (Gaussian binomial form) } P_n(q) = q^{-(m-1)\binom{n}{2}} \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q, \text{ and } R_n(q) = \sum_{d \mid n} \mu(n/d) P_d(q^{n/d}).$$

- (2) (*Root-of-unity collapse*) For every ζ with $\zeta^n = 1$: $R_n(\zeta) = n \sum_{C \in U_n^{\text{prim}}/C_n} \zeta^{\text{wt}(C)}$.
- (3) (*Derivative theorem*) The derivatives of the Laurent polynomials P_n and R_n vanish at every n -th root of unity. Equivalently, for $B = \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$ and every $\zeta^n = 1$,

$$\zeta B'(\zeta) = (m-1) \binom{n}{2} B(\zeta).$$

- (4) (*q -supercongruence*) For every odd prime $p \mid n$,

$$R_n(q) \equiv 0 \pmod{\Phi_p(q)^2}$$

in the localization $\mathbb{Z}[q^{\pm 1}]$, Φ_p the p -th cyclotomic polynomial. For $p = 2 \mid n$: if m is odd or $n \equiv 0 \pmod{4}$ then $R_n \equiv 0 \pmod{\Phi_2^2}$, while if m is even and $n \equiv 2 \pmod{4}$ then $R_n(-1) = -n \bar{Q}_{n/2}^{\text{prim}}(1)$ and $R'_n(-1) = 0$.

(Parts (1)–(4) are proved below as Propositions 4.1, 4.2 and Theorems 4.3, 4.5.)

Part (4) is, to our knowledge, the first q -supercongruence statement attached to Donaldson–Thomas invariants of any flavor. Note that $R_n(1) = n \bar{Q}_n^{\text{prim}}(1)$, which equals $n^2 \text{DT}_n$ when m is odd or $n \not\equiv 2 \pmod{4}$ (and $n^2 \text{DT}_n - n \bar{Q}_{n/2}^{\text{prim}}(1)$ otherwise); thus Theorem 1.5(4) is a genuine q -lift of the divisibility $n^2 \mid R_n(1)$, while Theorem 1.2 measures the deeper p -adic layers at $q = 1$.

Finally we sharpen Reineke’s congruence $\bar{Q}_n \equiv \frac{1}{n} \sum_{d \mid n} \mu(n/d) P_d(q^{n/d}) \pmod{q^n - 1}$ to an exact plethystic formula. Call $a \in U_n$ *admissible* if $\sum_{j \leq i} a_j \leq (m-1)i$ for all i , and let $A_n(q) = \sum_{a \text{ admissible}} q^{\text{wt}(a)}$, $A(q, t) = 1 + \sum_{n \geq 1} A_n(q) t^n$; here $A_n(1) = \frac{1}{(m-1)n+1} \binom{mn}{n}$ is the Fuss–Catalan number and A_n is a (co-area) q -Fuss–Catalan polynomial. Let Exp/Log denote the plethystic exponential/logarithm for the operations $\psi_r(q) = q^r$, $\psi_r(t) = t^r$.

Theorem 1.6 (Exact necklace formula). *For all $m \geq 2$, $n \geq 1$,*

$$\bar{Q}_n(q) = [t^n] \text{Log } A(q, t) + \begin{cases} \psi_2([t^{n/2}] \text{Log } A(q, t)), & m \text{ even and } n \equiv 2 \pmod{4}, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, $A(q, t) = \text{Exp}(\sum_{n \geq 1} \bar{Q}_n^{\text{prim}}(q) t^n)$ where $\bar{Q}_n^{\text{prim}}$ is the primitive-orbit part of $\bar{Q}_n$.

Combined with (5) this gives a closed recursive determination of all $\text{DT}_n^{(m)}(q)$ from ballot-path polynomials, and explains structurally why the arithmetic of quantized DT invariants is governed by (q -)Fuss–Catalan combinatorics.

1.4. Phenomenology modulo Φ_p , and conjectures. Section 7 reports exact computations of $\text{DT}_n^{(m)}(q) \pmod{\Phi_p(q)}$ over large ranges. The normalized values $V_n := q^{1-n} \text{DT}_n(q) \pmod{\Phi_p}$ display striking regularities — e.g. for $m = 2$, $p = 5$ the rows $n \equiv 2, 3, 4 \pmod{5}$ recur with integer multiplier sequences $(1, 1, 3, 9, \dots)$, $(1, 2, 6, 17, \dots)$, and the rows $n \equiv 1 \pmod{5}$ vanish except when $n \equiv 6 \pmod{20}$, where the doubling classes contribute — which we record as data and conjecture to be governed by the algebraicity of $A(\zeta_p, t)$ (Proposition 5.5). These finer questions are the natural q -analogues of the Jacobsthal layer and are left open, echoing the open problem of Straub [St11, Rem. 6] on a q -Jacobsthal congruence.

1.5. Verification. All statements have been verified by two independent implementations: a first-principles CoHA/plethystic engine computing $\text{DT}_n^{(m)}(q)$ from (2) (for $m = 2$ up to $n = 50$; $m = 3$: 36; $m = 4$: 27; $m = 5$: 22; $m = 6$: 14), and a clean-room enumeration of the cyclic-word model (5). The two agree on all overlapping ranges, and match the published tables of [Re12, KRSS19]. Theorem 1.2 was additionally checked directly from (3) for all $m \leq 10$, $n \leq 120$, and the tower congruences (4) on wide ranges. Section 8 gives details.

Related work. Apart from the figure-eight case treated by Basor–Conrey–Morrison [BCM17] (discussed above), arithmetic divisibility properties of DT-type invariants appear not to have been studied; the nearest circles of ideas are the 2-function literature [KSV06, SVW17, Mü21], congruence skein relations [CLPZ23], and the Habiro-ring arithmetic of Kontsevich–Soibelman series [GSWZ24]. On the number-theoretic side our tools come from the Wolstenholme–Ljunggren–Jacobsthal–Kazandzidis tradition [Me11, RZ95] and, for the q -story, we place our derivative theorem alongside the q -congruence results of Andrews, Clark, Straub, Pan, Zudilin and Gorodetsky [An99, Cl95, St11, St19, Pan13, Zu19, Go19], and the necklace-model Gauss-congruence framework of Gossow [Gos24]. Theorem 1.5(3) appears to be new even as a statement about Gaussian binomial coefficients, though we note that Gorodetsky’s root-of-unity derivative evaluations [Go19, Cor. 8.2] provide an alternative route to such statements.

Acknowledgments. The results in this paper were found and proved with essential help from computer experimentation; the search strategy, proofs and text are due to the author. Complete source code for all computations accompanies the paper.

2. CONVENTIONS AND BACKGROUND

Convention 2.1. Throughout, $m \geq 2$ is fixed and Q_m is the quiver with one vertex and m loops, with Euler form $\chi(a, b) = (1-m)ab$. We use Reineke’s normalization (1)–(2) for $\text{DT}_n^{(m)}(q)$. The dictionary to other conventions is: Efimov’s invariants [Ef12, §4] are obtained by $q \mapsto q^{-1}$ and rescaling $t \mapsto (-1)^{m-1}q^{(1-m)/2}t$; the Kontsevich–Soibelman invariants $\Omega_n(q^{1/2})$ of [KS11, §2.5] satisfy $\Omega_n(q) = q^{-(m-1)n/2}\text{DT}_n^{(m)}(q^{-1})$ up to the global sign $(-1)^{(m-1)n}$. Under either dictionary, Efimov’s theorem [Ef12] states: $\text{DT}_n^{(m)}(q) \in \mathbb{N}[q]$ is monic of degree $(m-1)\binom{n}{2}$ and divisible by q^{n-1} . All our computations confirm this normalization (Section 8).

Remark 2.2 (Erratum in [Re12, Thm. 6.8]). Theorem 6.8 of [Re12] is printed with the prefactor q^{1-n} , i.e. $\text{DT}_n(q) = q^{1-n}\bar{Q}_n(q)/[n]_q$; the same form is quoted in [KS16, eq. (B.4)]. This is inconsistent with the paper’s own Definition (via (2)) and with the degree/divisibility constraints of [Ef12, Cor. 4.2]: comparing the Exp-forms termwise gives $\frac{\text{DT}_n(q)}{1-q^{-1}} = \frac{\bar{Q}_n(q)}{1-q^{-n}}$, i.e. the correct prefactor is q^{n-1} as in (5). (For example, $m = 2, n = 2$: $\bar{Q}_2 = 1 + q$ — the primitive class $\{(2, 0), (0, 2)\}$ contributes q and the doubling class $(1, 1)$ contributes 1 — and $q^{n-1}\bar{Q}_2/[2]_q = q$ matches $\text{DT}_2^{(2)}(q) = q$, whereas q^{1-n} gives $q^{-1} \notin \mathbb{N}[q]$.) We verified (5) independently by symbolic expansion of (2) through t^{12} for $m = 2$ and t^9 for $m = 3$ and by the CoHA engine on the full ranges above. At $q = 1$ the two versions agree, so no numerical statement in [Re12] is affected.

We will use the Möbius function μ , the q -integer $[n]_q$, the Gaussian binomials $\begin{bmatrix} n \\ k \end{bmatrix}_q \in \mathbb{N}[q]$, and the cyclotomic polynomials $\Phi_d(q)$, with $q^n - 1 = \prod_{d|n} \Phi_d(q)$ and $[n]_q = \prod_{1 \leq d|n} \Phi_d(q)$. Recall that $\begin{bmatrix} a+b \\ b \end{bmatrix}_q$ is the generating polynomial of partitions in a $b \times a$ box.

Our arithmetic input is the following classical family of supercongruences. For the history — Babbage, Wolstenholme, Glaisher, Ljunggren, Jacobsthal — we refer to Meštrović’s survey [Me11].

Theorem 2.3 (Kazandzidis [Kaz68]; Robert–Zuber [RZ95]). *Let $n \geq k \geq 1$. Then*

$$\binom{pn}{pk} \equiv \binom{n}{k} \pmod{p^3 nk(n-k) \binom{n}{k} \mathbb{Z}_p} \quad (p \geq 5),$$

$$\binom{3n}{3k} \equiv \binom{n}{k} \pmod{3^2 nk(n-k) \binom{n}{k} \mathbb{Z}_3}.$$

These two statements are exactly as printed in [RZ95], which gives a short proof via the p -adic Morita gamma function (see also [Ro00, VII.1.6]). At the prime 2 the analogous congruence acquires a sign; since we could not locate an accessible published proof of this case (Kazandzidis's original articles [Kaz68] are hard to obtain), we give a complete elementary proof in Appendix B.

Proposition 2.4 (Signed Kazandzidis congruence at $p = 2$). *For all $n \geq k \geq 1$,*

$$\binom{2n}{2k} \equiv (-1)^{k(n-k)} \binom{n}{k} \pmod{2nk(n-k) \binom{n}{k} \mathbb{Z}_2}.$$

(We verified Theorem 2.3 computationally for all $2 \leq n \leq 40$ and Proposition 2.4 for all $2 \leq n \leq 300$, $1 \leq k < n$; the constants 3, 2, 1 are sharp, e.g. the $p = 2$ congruence is exact at $(n, k) = (2, 1)$: $\binom{4}{2} - (-1)\binom{2}{1} = 8$.)

3. THE WEIGHT STATISTIC: THREE LEMMAS

Recall $U_n = \{a \in \mathbb{N}^n : \sum_i a_i = (m-1)n\}$, $\text{wt}(a) = \sum_{i=1}^n (n-i)(m-1-a_i)$, the rotation $\rho(a) = (a_2, \dots, a_n, a_1)$, and the partial sums $S_k(a) = a_1 + \dots + a_k$ ($S_0 = 0$). Note the two elementary normalizations

$$(6) \quad \sum_{i=1}^n (m-1-a_i) = 0, \quad \text{wt}(a) = (m-1)\binom{n}{2} - \sum_{i=1}^n (n-i)a_i.$$

Lemma 3.1 (Rotation). *For every $a \in U_n$: $\text{wt}(\rho a) = \text{wt}(a) + n(a_1 - (m-1))$. Consequently*

$$\text{wt}(\rho^k a) = \text{wt}(a) + n(S_k(a) - (m-1)k) \quad (0 \leq k \leq n),$$

and wt is constant modulo n on each C_n -orbit.

Proof. Since $(\rho a)_i = a_{i+1}$ for $i < n$ and $(\rho a)_n = a_1$ (whose weight-coefficient is $n - n = 0$),

$$\text{wt}(\rho a) = \sum_{j=2}^n (n-j+1)(m-1-a_j) = \sum_{j=2}^n (n-j)(m-1-a_j) + \sum_{j=2}^n (m-1-a_j).$$

The first sum is $\text{wt}(a) - (n-1)(m-1-a_1)$; by (6) the second is $-(m-1-a_1)$. Adding gives $\text{wt}(a) - n(m-1-a_1)$. The iterated form follows by induction, and $\text{wt}(\rho^k a) \equiv \text{wt}(a) \pmod{n}$. $\square$

Lemma 3.2 (Repetition). *Let $b \in U_d$ and let $b^{(r)} \in U_{rd}$ be its r -fold concatenation. Then $\text{wt}_{rd}(b^{(r)}) = r \text{wt}_d(b)$.*

Proof. Write $n = rd$. Then

$$\text{wt}_n(b^{(r)}) = \sum_{t=0}^{r-1} \sum_{j=1}^d (n - td - j)(m-1-b_j) = \sum_{t=0}^{r-1} \left[\sum_{j=1}^d (d-j)(m-1-b_j) + (n-td-d) \sum_{j=1}^d (m-1-b_j) \right],$$

and the inner second sum vanishes by (6) applied to $b \in U_d$; each bracket equals $\text{wt}_d(b)$. $\square$

Lemma 3.3 (Orbit-sum identity). *For every $a \in U_n$: $\sum_{k=0}^{n-1} \text{wt}(\rho^k a) = 0$.*

Proof. By Lemma 3.1, the sum equals $n \text{wt}(a) + n(\sum_{k=1}^{n-1} S_k(a) - (m-1)\binom{n}{2})$. Now

$$\sum_{k=1}^{n-1} S_k(a) = \sum_{i=1}^n a_i \# \{k : i \leq k \leq n-1\} = \sum_{i=1}^n (n-i)a_i \stackrel{(6)}{=} (m-1)\binom{n}{2} - \text{wt}(a),$$

so the total is $n \text{wt}(a) + n[(m-1)\binom{n}{2} - \text{wt}(a) - (m-1)\binom{n}{2}] = 0$. $\square$

If a has period $d \mid n$ (i.e. $\rho^d a = a$), then $\sum_{k=0}^{n-1} \text{wt}(\rho^k a) = \frac{n}{d} \sum_{k=0}^{d-1} \text{wt}(\rho^k a)$, so *the orbit-sum of wt over each C_n -orbit (of any period) vanishes*.

Finally, call $a \in U_n$ *admissible* if $S_k(a) \leq (m-1)k$ for all $0 \leq k \leq n$. By Lemma 3.1, $\text{wt}(\rho^k a) \leq \text{wt}(a)$ for all k iff a is admissible; since orbits are finite, every orbit contains an admissible element, and the admissible elements of an orbit are exactly the rotations attaining the maximal weight $\text{wt}(C)$ (this is Lemma 6.5 of [Re12], which we have just reproved).

4. EXACT IDENTITIES AND THE DERIVATIVE THEOREM

Proposition 4.1. $P_n(q) := \sum_{a \in U_n} q^{\text{wt}(a)} = q^{-(m-1)\binom{n}{2}} \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$, *a symmetric ($q \leftrightarrow q^{-1}$ -invariant) Laurent polynomial supported in $[-(m-1)\binom{n}{2}, (m-1)\binom{n}{2}]$.*

Proof. By (6), $P_n(q) = q^{(m-1)\binom{n}{2}} \sum_a q^{-\sum_i (n-i)a_i}$, the sum over compositions of $N := (m-1)n$ into n parts $a_1, \dots, a_n \geq 0$ with part a_i weighted $(n-i)a_i$. Mapping a to the partition with a_i parts of size $n-i$ gives a bijection onto partitions with at most N parts, each of size $\leq n-1$ — i.e. partitions in an $N \times (n-1)$ box — transforming the statistic into the partition size. Hence $\sum_a x^{\sum_i (n-i)a_i} = \left[\begin{matrix} N+n-1 \\ n-1 \end{matrix} \right]_q \Big|_{q=x}$, and with $N+n-1 = mn-1$,

$$P_n(q) = q^{(m-1)\binom{n}{2}} \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q \Big|_{q \rightarrow q^{-1}} = q^{(m-1)\binom{n}{2} - \deg} \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$$

using the palindromicity of Gaussian binomials, where $\deg = (mn-1-(n-1))(n-1) = 2(m-1)\binom{n}{2}$. $\square$

Proposition 4.2. *Let $R_n(q) := \sum_{a \in U_n^{\text{prim}}} q^{\text{wt}(a)}$. Then:*

- (1) $R_n(q) = \sum_{d \mid n} \mu(n/d) P_d(q^{n/d})$.
- (2) *For every $\zeta \in \mathbb{C}$ with $\zeta^n = 1$: $P_n(\zeta) = \sum_{d \mid n} d \sum_{C \in U_d^{\text{prim}}/C_d} \zeta^{\frac{n}{d} \text{wt}(C)}$ and $R_n(\zeta) = n \sum_{C \in U_n^{\text{prim}}/C_n} \zeta^{\text{wt}(C)}$, where $\zeta^{\text{wt}(C)}$ is well defined by Lemma 3.1.*

Proof. (1) Every $a \in U_n$ of exact period d is uniquely $b^{(n/d)}$ with $b \in U_d^{\text{prim}}$, and $\text{wt}(a) = \frac{n}{d} \text{wt}_d(b)$ by Lemma 3.2. Hence $P_n(q) = \sum_{d \mid n} R_d(q^{n/d})$, and Möbius inversion (in the Dirichlet sense, using $\psi_r \psi_s = \psi_{rs}$ for the substitutions $q \mapsto q^r$) gives (1). (2) Group U_n^{prim} into orbits: each orbit has n elements whose weights differ by multiples of n (Lemma 3.1), so all contribute $\zeta^{\text{wt}(C)}$; this gives the formula for $R_n(\zeta)$, and the one for $P_n(\zeta)$ follows from the period decomposition together with Lemma 3.2. $\square$

Theorem 4.3 (Derivative theorem). *For every $n \geq 1$ and every $\zeta \in \mathbb{C}$ with $\zeta^n = 1$,*

$$P'_n(\zeta) = 0 \quad \text{and} \quad R'_n(\zeta) = 0,$$

where ' denotes $\frac{d}{dq}$ of Laurent polynomials. Equivalently, with $B = \begin{bmatrix} mn-1 \\ n-1 \end{bmatrix}_q$ and $s = (m-1)\binom{n}{2}$:

$$\zeta B'(\zeta) = sB(\zeta) \quad \text{for every } \zeta^n = 1.$$

Proof. Fix ζ with $\zeta^n = 1$. Group the sum $R'_n(q) = \sum_{a \in U_n^{\text{prim}}} \text{wt}(a) q^{\text{wt}(a)-1}$ into C_n -orbits. On an orbit C with representative a , all exponents $\text{wt}(\rho^k a) - 1$ are congruent mod n (Lemma 3.1), so

$$\sum_{k=0}^{n-1} \text{wt}(\rho^k a) \zeta^{\text{wt}(\rho^k a)-1} = \zeta^{\text{wt}(C)-1} \sum_{k=0}^{n-1} \text{wt}(\rho^k a) = 0$$

by Lemma 3.3. Summing over orbits gives $R'_n(\zeta) = 0$. For P_n , orbits of period $d \mid n$ contribute $\zeta^{\text{wt}(C)-1} \sum_{k=0}^{d-1} \text{wt}(\rho^k a)$, which vanishes by the remark after Lemma 3.3. Finally $P_n = q^{-s}B$ gives $P'_n = q^{-s-1}(qB' - sB)$, so $P'_n(\zeta) = 0$ is equivalent to $\zeta B'(\zeta) = sB(\zeta)$. $\square$

Remark 4.4. At $\zeta = 1$ the identity $B'(1) = sB(1)$ is the palindromy of B (the mean of the size statistic over the $N \times (n-1)$ box is s); at the other n -th roots of unity the statement appears to be new. It is a genuinely different family from the evaluations $\omega \frac{d}{dq} \begin{bmatrix} an \\ bn \end{bmatrix}_q(\omega)$ computed by Gorodetsky [Go19, Thm. 2.6], which are in general nonzero. Note also that Theorem 4.3 may be restated as: $(q^n - 1) \mid (q\hat{R}'_n(q) - \sigma\hat{R}_n(q))$ in $\mathbb{Z}[q]$, where $\hat{R}_n = q^\sigma R_n$ is the polynomial normalization; this is the form in which it was machine-verified on large ranges.

Theorem 4.5 (q -supercongruences for the primitive part). *Let $p \mid n$ be prime.*

- (1) *If p is odd, then $\Phi_p(q)^2$ divides $R_n(q)$ in $\mathbb{Z}[q^{\pm 1}]$.*
- (2) *If $p = 2$: when m is odd or $4 \mid n$, $\Phi_2(q)^2 = (q+1)^2$ divides $R_n(q)$; when m is even and $n \equiv 2 \pmod{4}$,*

$$R_n(-1) = -n\bar{Q}_{n/2}^{\text{prim}}(1), \quad R'_n(-1) = 0.$$

Proof. By Theorem 4.3, R'_n vanishes at every primitive p -th root of unity ζ_p (these satisfy $\zeta_p^n = 1$ since $p \mid n$). It remains to evaluate $R_n(\zeta_p)$. By Proposition 4.2(2), $R_n(\zeta_p) = n \sum_C \zeta_p^{\text{wt}(C)}$ over primitive orbits. By (5) and $[n]_q \mid \bar{Q}_n$ [Re12, Thm. 6.7], and since $\Phi_p \mid [n]_q$, we get $\bar{Q}_n(\zeta_p) = 0$. If m is odd or $n \not\equiv 2 \pmod{4}$ then $\bar{Q}_n = \bar{Q}_n^{\text{prim}} = \sum_C q^{\text{wt}(C)}$, whence $R_n(\zeta_p) = n\bar{Q}_n(\zeta_p) = 0$; double vanishing gives $\Phi_p^2 \mid R_n$. If m is even and $n \equiv 2 \pmod{4}$ then $\bar{Q}_n = \bar{Q}_n^{\text{prim}} + \psi_2(\bar{Q}_{n/2}^{\text{prim}})$ (Theorem 1.6, or [Re12, Thm. 5.11] with Lemma 3.2), so

$$R_n(\zeta_p) = n\bar{Q}_n^{\text{prim}}(\zeta_p) = n(\bar{Q}_n(\zeta_p) - \bar{Q}_{n/2}^{\text{prim}}(\zeta_p^2)) = -n\bar{Q}_{n/2}^{\text{prim}}(\zeta_p^2).$$

For odd p , ζ_p^2 is again a primitive p -th root of unity and $p \mid \frac{n}{2}$; the same argument one level down (note $\frac{n}{2}$ is odd, so no further doubling) gives $\bar{Q}_{n/2}^{\text{prim}}(\zeta_p^2) = \bar{Q}_{n/2}(\zeta_p^2) = 0$. For $p = 2$, $\zeta_2 = -1$ and $\zeta_2^2 = 1$, giving $R_n(-1) = -n\bar{Q}_{n/2}^{\text{prim}}(1)$. $\square$

Remark 4.6. $\bar{Q}_{n/2}^{\text{prim}}(1) = \#(U_{n/2}^{\text{prim}}/C_{n/2}) = \frac{2}{n} \sum_{d \mid n/2} \mu\left(\frac{n}{2d}\right) \binom{md-1}{d-1}$ is explicit; so part (2) is an exact evaluation, not merely an obstruction. Note also that all statements descend to every ζ of order $p^j \mid n$, Φ_{p^j} in place of Φ_p , by the same proof.

5. THE EXACT NECKLACE FORMULA

We now prove Theorem 1.6. Recall $A_n(q) = \sum_{a \in U_n \text{ admissible}} q^{\text{wt}(a)}$ and $A(q, t) = 1 + \sum_{n \geq 1} A_n t^n$. On power series with coefficients in $\mathbb{Z}[q^{\pm 1}]$ and no constant-term contribution we use the plethystic operations determined by $\psi_r(q^w t^l) = q^{rw} t^{rl}$, with

$$\text{Exp}(F) = \exp\left(\sum_{r \geq 1} \frac{\psi_r(F)}{r}\right), \quad \text{Log} = \text{Exp}^{-1}, \quad \text{Exp}(q^w t^l) = \frac{1}{1 - q^w t^l}.$$

Definition 5.1. A *block* is an admissible $c \in U_j$, $j \geq 1$, whose partial sums satisfy $S_k(c) < (m-1)k$ for $0 < k < j$ (no interior return to the diagonal). Let $\mathfrak{B}$ denote the set of blocks (of all lengths $j \geq 1$), a graded alphabet with finitely many letters in each degree, and give the letter c the monomial weight $u_c = q^{\text{wt}(c)} t^{|c|}$.

Lemma 5.2 (First-return factorization; additivity). *Concatenation gives a bijection between words $c^{(1)}c^{(2)} \dots c^{(r)}$ in the free monoid $\mathfrak{B}^*$ and admissible sequences, under which lengths add and weights add: if $u \in U_j$ satisfies $S_j(u) = (m-1)j$ (in particular if u is a block or a concatenation of blocks) and $v \in U_l$, then $\text{wt}_{j+l}(uv) = \text{wt}_j(u) + \text{wt}_l(v)$. Consequently*

$$A(q, t) = \frac{1}{1 - B(q, t)}, \quad B(q, t) := \sum_{c \in \mathfrak{B}} u_c.$$

Proof. An admissible $a \in U_n$ returns to the diagonal ($S_k = (m-1)k$) at a set of indices $0 = k_0 < k_1 < \dots < k_r = n$; cutting at these indices expresses a uniquely as a concatenation of blocks, and conversely any concatenation of blocks is admissible with returns exactly at the cut points. For additivity, write $n = j + l$:

$$\text{wt}_n(uv) = \sum_{i \leq j} (n-i)(m-1-u_i) + \sum_{i' \leq l} (l-i')(m-1-v_{i'})$$

where the first sum equals $\sum_{i \leq j} (j-i)(m-1-u_i) + (n-j) \sum_{i \leq j} (m-1-u_i) = \text{wt}_j(u) + 0$ by $\sum_{i \leq j} (m-1-u_i) = 0$ (which holds as $S_j(u) = (m-1)j$), and the second sum is $\text{wt}_l(v)$ on the nose. The generating identity follows since weights and lengths are additive. $\square$

Lemma 5.3 (Classes vs. necklaces). *Cutting at diagonal returns induces a bijection*

$$U_n^{\text{prim}}/C_n \longleftrightarrow \{\text{aperiodic necklaces over } \mathfrak{B} \text{ of total length } n\},$$

under which $\text{wt}(C)$ equals the total block weight of the necklace. Moreover the admissible elements in a primitive class C are exactly its block-boundary rotations; there are r_C of them (r_C = number of blocks), all of weight $\text{wt}(C)$.

Proof. Let C be a primitive class and $a \in C$ admissible (Section 3). By Lemma 3.1, a rotation $\rho^k a$ has weight $\text{wt}(a)$ iff $S_k(a) = (m-1)k$, i.e. iff k is a return; and being admissible is equivalent to having maximal weight, so the admissible elements of C are exactly the rotations at returns, i.e. the block-boundary rotations of the block word of a ; their number is the number of blocks r_C . The block word is determined by C up to cyclic rotation, giving a well-defined necklace; it is aperiodic, since a period of the block word is in particular a period of a . Conversely an aperiodic necklace, read as a sequence, is primitive: a sequence-period $d \mid n$, $d < n$, would make $\text{wt}(\rho^d a) = \text{wt}(a)$ maximal, hence $\rho^d a$ admissible with the same block structure shifted, exhibiting a period of the necklace. The weight statement is Lemma 5.2. Both constructions are mutually inverse. $\square$

Proof of Theorem 1.6. Fix any total order on the countable alphabet $\mathfrak{B}$. By the Chen–Fox–Lyndon theorem, every word in $\mathfrak{B}^*$ factors uniquely as a weakly decreasing concatenation of

Lyndon words; since length and weight are additive (Lemma 5.2), this factorization gives an isomorphism of graded monoids between $\mathfrak{B}^*$ and the free commutative monoid on Lyndon words, whence the generating identity

$$A(q, t) = \frac{1}{1 - B(q, t)} = \prod_{\ell \text{ Lyndon}} \frac{1}{1 - u_\ell} = \text{Exp} \left(\sum_{\ell \text{ Lyndon}} u_\ell \right),$$

using $\text{Exp}(u) = 1/(1 - u)$ for each monomial u_ℓ and multiplicativity of Exp ; all products/sums are locally finite since each t -degree contains finitely many letters. Lyndon words over $\mathfrak{B}$ biject with aperiodic necklaces, so by Lemma 5.3,

$$\sum_{\ell} u_\ell = \sum_{n \geq 1} \left(\sum_{C \in U_n^{\text{prim}}/C_n} q^{\text{wt}(C)} \right) t^n = \sum_{n \geq 1} \bar{Q}_n^{\text{prim}}(q) t^n.$$

Applying Log gives $\bar{Q}_n^{\text{prim}} = [t^n] \text{Log } A(q, t)$. Finally, by [Re12, Thm. 5.11 and §6], $\bar{Q}_n = \bar{Q}_n^{\text{prim}}$ unless m is even and $n \equiv 2 \pmod{4}$, in which case $\bar{Q}_n = \bar{Q}_n^{\text{prim}} + \sum_b q^{\text{wt}_n(bb)}$, the sum over primitive classes of $U_{n/2}$; by Lemma 3.2, $\text{wt}_n(bb) = 2\text{wt}_{n/2}(b)$ and (arguing as in Lemma 5.3 one level down, or directly since doubling preserves maximality) the class-maximum doubles, so the correction equals $\psi_2(\bar{Q}_{n/2}^{\text{prim}})$. $\square$

Remark 5.4. At $q = 1$, $A(1, t)$ is the Fuss–Catalan generating function ($A_n(1) = \frac{1}{(m-1)n+1} \binom{mn}{n}$, the ballot-number count of admissible sequences), and Theorem 1.6 degenerates to the classical necklace form of (3). The polynomials $A_n(q)$ are the images of the Poincaré polynomials of the noncommutative Hilbert schemes $\text{Hilb}_{n,1}^{(m)}$ of [Re05] under $q \mapsto q^{-1}$ up to a monomial (co-area vs. area statistic); we do not use this here. We also stress how Theorem 1.6 differs from Reineke’s exact Exp -identity [Re12, Thm. 5.11] combined with his framed factorization [Re12, Cor. 2.2(4)]: those identities are formulated for the area-statistic (Poincaré-polynomial) generating series, and the naive area-statistic analogue of Theorem 1.6 is *false* (we checked that it fails already at $n = 3$); it is precisely in the co-area gauge that the plethystic logarithm of the ballot series produces exactly $\bar{Q}_n$.

Proposition 5.5 (Algebraicity at roots of unity). *Let ζ be a root of unity. Then $A(\zeta, t)$ is algebraic over $\mathbb{Q}(\zeta)(t)$.*

Proof. Let $F(q, t) = \sum_n F_n(q) t^n$ be the area generating function of admissible sequences, i.e. $F_n(q) = \sum_a q^{(m-1)\binom{n}{2} - \text{wt}(a)}$ (sum over admissible $a \in U_n$), so that $A_n(q) = q^{(m-1)\binom{n}{2}} F_n(q^{-1})$ and $F_n \in \mathbb{N}[q]$ has degree $\leq (m-1)\binom{n}{2}$. By Reineke’s grafting recursion [Re12, Cor. 2.2(1)] (in the tree model of [Re05], which is equivalent to the ballot model by the standard bijection between m -ary trees and admissible lattice words), F satisfies the q -difference equation

$$F(q, t) = 1 + t \prod_{k=0}^{m-1} F(q, q^k t).$$

Let p be the order of ζ . Specializing $q = \zeta$ and substituting $t \mapsto \zeta^j t$ in the q -difference equation, the finitely many series $G_j(t) := F(\zeta, \zeta^j t)$, $j \in \mathbb{Z}/p$, satisfy the polynomial system

$$G_j(t) = 1 + \zeta^j t \prod_{k=0}^{m-1} G_{j+k}(t) \quad (j \in \mathbb{Z}/p)$$

— a finite system of polynomial equations over $\mathbb{Q}(\zeta)(t)$ with the formal-power-series solution $(G_j)_j$. The Jacobian of the system $\Psi_j := G_j - 1 - \zeta^j t \prod_k G_{j+k}$ with respect to $(G_j)_j$ is the identity matrix modulo t , hence invertible along the formal solution; the solution is therefore a smooth

isolated point of the fiber of the associated affine variety over the generic point of $\text{Spec } \mathbb{Q}(\zeta)[t]$, so it lies on a zero-dimensional component over $\mathbb{Q}(\zeta)(t)$ and its coordinates are algebraic over $\mathbb{Q}(\zeta)(t)$. Hence $F(\zeta, t) = G_0(t)$ is algebraic. Finally, $A(\zeta, t) = \sum_n \zeta^{(m-1)\binom{n}{2}} F_n(\zeta^{-1}) t^n$: the exponent $(m-1)\binom{n}{2}$ is (for fixed m, p) a quadratic function of $n \bmod p$, hence periodic in n of period dividing $2p$; therefore $A(\zeta, t)$ is a finite sum of arithmetic-progression sections of the algebraic series $F(\zeta^{-1}, t)$, each twisted by a constant. Sections along arithmetic progressions preserve algebraicity (averaging over the substitutions $t \mapsto \omega t$, $\omega^{2p} = 1$), so $A(\zeta, t)$ is algebraic. $\square$

This explains qualitatively the linear-recursive patterns of Section 7; making the resulting evaluations of $\text{DT}_n(\zeta_p)$ fully explicit is an interesting problem that we leave open.

6. PROOF OF THE IMPROVED INTEGRALITY THEOREM

Throughout this section put, for $e \geq 1$,

$$a_e := (-1)^{(m-1)(e-1)} \binom{me-1}{e-1} \in \mathbb{Z},$$

so that (3) reads

$$(7) \quad n^2 \text{DT}_n = (-1)^{(m-1)(n-1)} \sum_{d|n} \mu\left(\frac{n}{d}\right) a_d.$$

We use throughout the identity

$$(8) \quad \binom{me}{e} = m \binom{me-1}{e-1}.$$

Lemma 6.1 (Tower congruences). *Let p be prime, $e \geq 1$. Set $\kappa_p = 3$ for $p \geq 5$, $\kappa_3 = 2$, $\kappa_2 = 1$. Then*

$$v_p(a_{pe} - a_e) \geq \kappa_p + 3v_p(e) + v_p(m(m-1)).$$

Proof. Apply Theorem 2.3 (for $p \geq 3$), resp. Proposition 2.4 (for $p = 2$), with $n = me$, $k = e$:

$$\binom{mpe}{pe} \equiv \varepsilon \binom{me}{e} \pmod{p^{\kappa_p} me \cdot e \cdot (m-1)e \cdot \binom{me}{e} \mathbb{Z}_p},$$

where $\varepsilon = 1$ for p odd and $\varepsilon = (-1)^{e \cdot (m-1)e} = (-1)^{(m-1)e}$ for $p = 2$. The p -adic valuation of the modulus is at least

$$\kappa_p + v_p(m) + 3v_p(e) + v_p(m-1) + v_p\left(\binom{me}{e}\right) \geq \kappa_p + 3v_p(e) + v_p(m(m-1)) + v_p(m),$$

where we used $v_p\left(\binom{me}{e}\right) \geq v_p(m)$, immediate from (8). Dividing the congruence by m (using (8) on both sides) gives

$$\binom{mpe-1}{pe-1} \equiv \varepsilon \binom{me-1}{e-1} \pmod{p^{\kappa_p+3v_p(e)+v_p(m(m-1))} \mathbb{Z}_p}.$$

It remains to compare with the signs in a_e . For p odd, $(m-1)(pe-1) \equiv (m-1)(e-1) \pmod{2}$ since $pe \equiv e$, so a_{pe} and a_e carry the same sign and $\varepsilon = 1$: the displayed congruence is exactly the claim. For $p = 2$:

$$a_{2e} = (-1)^{(m-1)(2e-1)} \binom{2me-1}{2e-1} = (-1)^{m-1} \binom{2me-1}{2e-1},$$

and the displayed congruence gives $\binom{2me-1}{2e-1} \equiv (-1)^{(m-1)e} \binom{me-1}{e-1}$, so

$$a_{2e} \equiv (-1)^{(m-1)+(m-1)e+(m-1)(e-1)} a_e = (-1)^{2(m-1)e} a_e = a_e,$$

modulo the same power of 2. The signs cancel identically, completing the proof. $\square$

Remark 6.2. The exact cancellation of signs at $p = 2$ is the arithmetic reflection of the sign twist $(-1)^{(m-1)n}$ in (2): the motivic sign conventions of DT theory are precisely the ones compatible with the 2-adic Kazandzidis congruence.

Proof of Theorem 1.2. Fix a prime p and n with $v := v_p(n) \geq 1$; write $n = p^v u$, $p \nmid u$. In (7), every divisor of n is $d = p^j f$ with $f \mid u$, $0 \leq j \leq v$, and $\mu(n/d) = \mu(p^{v-j})\mu(u/f)$ vanishes unless $j \in \{v, v-1\}$. Hence

$$\sum_{d \mid n} \mu\left(\frac{n}{d}\right) a_d = \sum_{f \mid u} \mu\left(\frac{u}{f}\right) (a_{p^v f} - a_{p^{v-1} f}).$$

By Lemma 6.1 with $e = p^{v-1} f$,

$$v_p(a_{p^v f} - a_{p^{v-1} f}) \geq \kappa_p + 3(v-1) + v_p(m(m-1)),$$

so $v_p(n^2 \text{DT}_n) \geq 3v + (\kappa_p - 3) + v_p(m(m-1))$ and therefore

$$(9) \quad v_p(\text{DT}_n) \geq v + (\kappa_p - 3) + v_p(m(m-1)).$$

For $p \geq 5$, $\kappa_p - 3 = 0$ gives (1). For $p = 3$: (9) reads $v_3(\text{DT}_n) \geq v - 1 + v_3(m(m-1))$; if $m \equiv 0, 1 \pmod{3}$ then $3 \mid m(m-1)$ and we get $\geq v$; if $m \equiv 2$ we get $\geq v - 1$. This is (2). For $p = 2$: (9) reads $v_2(\text{DT}_n) \geq v - 2 + v_2(m(m-1))$. Exactly one of $m, m-1$ is even and $v_2(m(m-1))$ equals $v_2(m)$ or $v_2(m-1)$; it is ≥ 2 iff $m \equiv 0, 1 \pmod{4}$, and equals 1 iff $m \equiv 2, 3 \pmod{4}$. This is (3).

Optimality: by (3), $\text{DT}_2 = \lfloor m/2 \rfloor$ and $\text{DT}_3 = \binom{m}{2}$. If $m \equiv 2, 3 \pmod{4}$ then $\lfloor m/2 \rfloor$ is odd, so $v_2(\text{DT}_2) = 0 = v_2(2) - 1$. If $m \equiv 2 \pmod{3}$ then $3 \nmid m(m-1)$, so $v_3(\text{DT}_3) = 0 = v_3(3) - 1$. $\square$

Proof of Corollary 1.3. By [GKS16, Prop. 1.2] (transcribed in the introduction) the extremal BPS invariants of the twist knot K_p are $\pm \text{DT}_r^{(m)}$ for $m \in \{3, 2|p|+1\}$ ($p \leq -1$) and $m \in \{2, 2p+2\}$ ($p \geq 2$). Theorem 1.2 gives $r \mid \gamma(m) \text{DT}_r^{(m)}$ with the optimal $\gamma(m)$, which is Conjecture 1.1 with $\gamma^\pm = \gamma(m)$. Since $\gamma(m)$ depends only on $m \pmod{12}$, the values for the families $m = 2|p|+1$, $m = 2p+2$ depend only on $p \pmod{6}$, giving twelve values in total; we checked that they coincide with all twist-knot entries of [GKS16, Table 1]. For the figure-eight knot ($p = -1$): both rows have $m = 3$; $\varepsilon_2(3) = 1$, $\varepsilon_3(3) = 0$, so $\gamma = 2$, in accordance with [GKS16, Table 2] and [BCM17]. $\square$

Corollary 6.3 (Supercongruences). *For all primes $p \geq 5$ and all $u, k \geq 1$: $a_{p^k u} \equiv a_{p^{k-1} u} \pmod{p^{3k}}$. In the terminology of [AZ06, SVW17, Mü21], (a_e) (equivalently, the sequence $\binom{me-1}{e-1}$ up to sign) is a 3-sequence at every $p \geq 5$; at $p = 3$ and $p = 2$ it is a 3-sequence up to the fixed defects $3^{-1}, 2^{-2}$ improved by $v_p(m(m-1))$, as quantified in Lemma 6.1.*

Proof. Lemma 6.1 with $e = p^{k-1} u$. $\square$

7. PHENOMENOLOGY MODULO Φ_p AND OPEN PROBLEMS

Beyond Theorem 4.5, the invariants $\text{DT}_n(q)$ display rich structure at roots of unity. Table 1 shows, for $m = 2$ and $p = 5$, the normalized reductions $V_n := q^{1-n} \text{DT}_n(q) \pmod{\Phi_5(q)}$ (as vectors in the basis $1, q, q^2, q^3$), computed exactly from the engine data.

Several features are visible: (i) rows with $n \equiv 1 \pmod{5}$ vanish except exactly when $n \equiv 6 \pmod{20}$, i.e. when $n \equiv 2 \pmod{4}$, where by Theorem 4.5(2) the doubling classes contribute; (ii)

TABLE 1. $V_n = q^{1-n} \text{DT}_n^{(2)}(q) \bmod \Phi_5(q)$, $n \leq 30$.

$n \equiv 1$:	$V_1 = [1, 0, 0, 0]$, $V_6 = [-1, 0, 0, -1]$,	$V_{11} = V_{16} = V_{21} = 0$,	$V_{26} = [-1, 0, 0, -1]$
$n \equiv 2$:	$V_2 = [1, 0, 0, 0]$, $V_7 = 0$,	$V_{12} = [1, 0, 0, 0]$, $V_{17} = [1, 0, 0, 0]$,	$V_{22} = 3V_2$, $V_{27} = 9V_2$
$n \equiv 3$:	$V_3 = [0, 1, 0, 0]$, $V_8 = 0$,	$V_{13} = V_3$, $V_{18} = 2V_3$,	$V_{23} = 6V_3$, $V_{28} = 17V_3$
$n \equiv 4$:	$V_4 = [0, 1, 0, 1]$, $V_9 = 0$,	$V_{14} = V_4$, $V_{19} = 2V_4$,	$V_{24} = 6V_4$, $V_{29} = 17V_4$
$n \equiv 0$:	$V_5 = [-1, 1, 0, 0]$,	$V_{5k} = c_k \cdot [-1, 1, 0, 0]$,	$c_k = 1, 2, 4, 12, 41, 138$ ($k \leq 6$)

fixed nonzero residues recur with integer multiplier sequences $(1, 1, 3, 9, \dots)$ and $(1, 2, 6, 17, \dots)$; (iii) the diagonal column $n = 5k$ carries a single vector $[-1, 1, 0, 0] \propto (q - 1)$ with multipliers $1, 2, 4, 12, 41, 138, \dots$. Identical phenomena occur for other (m, p) . In view of Proposition 5.5 we expect all of this to be governed by the algebraic function $A(\zeta_5, t)$:

Conjecture 7.1. *For each m , each prime p and each residue b , the sequence of reductions $(V_{pk+b} \bmod \Phi_p)_{k \geq 0}$ lies, from some k on, in a one-dimensional lattice $\mathbb{Z} \cdot w_b$, and the resulting integer multiplier sequences are coefficient sequences of algebraic generating functions (explicitly computable from $A(\zeta_p, t)$ via Theorem 1.6).*

A second natural problem is the q -Jacobsthal layer: Theorem 4.5 is a Φ_p^2 -statement for R_n ; the numerical Theorem 1.2 shows that at $q = 1$ a third layer appears for $p \geq 5$. In analogy with Straub's question [St11, Rem. 6] we ask: is there a natural completion of Theorem 4.3 at the level of second derivatives — equivalently, an exact evaluation of $\sum_C (\sum_k e_k(C)^2) \zeta^{\text{wt}(C)}$, where $e_k(C)$ are the weight-drops along the orbit — from which the full p^3 -divisibility of Theorem 1.2 would follow q -analytically? Our data suggest this second-moment generating function is again Gaussian-binomial-like; we have not found its closed form.

8. COMPUTATIONAL VERIFICATION

Two independent implementations were written.

(a) *CoHA engine.* A first-principles implementation of the plethystic factorization (2): the Poincaré series of the (shifted) CoHA of Q_m is expanded in exact integer arithmetic (FLINT), the plethystic logarithm is extracted, and $\text{DT}_n^{(m)}(q)$ is read off. Truncation is made provably exact by a gauge trick: the t^n -coefficient is stored multiplied by $q^{+(m-1)n^2}$ -normalizations under which every algebraic operation (products of graded pieces, Adams operations) moves q -support upward only, so a global cap loses nothing below it. The engine reproduces: Efimov's constraints (monicity, degree $(m-1)\binom{n}{2}$, divisibility by q^{n-1} , nonnegativity) for all computed invariants; the published tables of [Re12] and (after the conversion of Convention 2.1) of [KRSS19]; and the anchor collapses for $m = 0, 1$ (a single fermionic, resp. bosonic, generator). Ranges: $m = 2$, $n \leq 50$; $m = 3$, $n \leq 36$; $m = 4$, $n \leq 27$; $m = 5$, $n \leq 22$; $m = 6$, $n \leq 14$.

(b) *Word model.* A second, structurally independent implementation: direct enumeration of $U_n^{\text{prim},+}/C_n$ with the weight statistic, implementing (5) literally ($m = 2$, $n \leq 12$; $m = 3$, $n \leq 9$; $m = 4$, $n \leq 7$; $m = 5, 6$, $n \leq 6$; up to 112,632 classes per invariant). It shares no code, no data structures and no algorithmic steps with the engine (a); in particular the engine never enumerates words and the word model never manipulates power series. The two implementations agree on all overlapping (m, n) , and the exactness of the division by $[n]_q$ in (5) was confirmed in every case.

(c) *Identity checks.* Theorem 1.6 was verified as an identity of polynomials for $m = 2, 3, 4$ up to $n = 10, 8, 8$; Proposition 4.1 for $m \leq 5$, $n \leq 9$ by direct enumeration against the Gaussian binomial; Theorem 4.3 in the polynomial form $(q^n - 1) \mid (q\hat{R}'_n - \sigma\hat{R}_n)$ for $m \leq 5$, $n \leq 10$; Lemmas 3.1–3.3 on $> 10^4$ random instances.

(d) *Arithmetic checks.* Theorem 1.2 was verified from (3) for all $2 \leq m \leq 10$, $n \leq 120$, all $p \leq 13$ (no failures; defects attained exactly on the predicted residue classes of m). Lemma 6.1 was verified for $m \leq 10$, $p \in \{2, 3, 5, 7\}$, $e \leq 50$ including high prime-power parts. Theorem 2.3 was verified for $2 \leq n \leq 40$, $1 \leq k < n$, $p \in \{3, 5, 7\}$, and Proposition 2.4 for $2 \leq n \leq 300$, with sharpness of the exponents 3, 2, 1; Theorem 4.5 was additionally verified end-to-end (including prime powers Φ_{p^j} and the exact $p = 2$ defect) for $m \leq 5$, $n \leq 18$.

(e) *Lean formalization.* The arithmetic core of the paper has been formalized in Lean 4 over Mathlib (file `Dtformal.lean`, included with the sources; it compiles with no `sorry`, and every statement depends only on Lean's three standard axioms). Machine-checked *unconditionally*: the identity $\binom{me}{e} = m \binom{me-1}{e-1}$; the Möbius collapse of Section 6 (in the form $S(p^v u) = \sum_{f|u} \mu(u/f) (a_{p^v f} - a_{p^{v-1} f})$ for $p \nmid u$, where $S(n) := \sum_{d|n} \mu(n/d) a_d$); the evaluations $S(2)$, $2S(3) = 9m(m-1)$ with the divisibilities $4 \mid S(2)$, $9 \mid S(3)$ and the sharpness statements $8 \nmid S(2)$ for $m \equiv 2, 3 \pmod{4}$ and $27 \nmid S(3)$ for $m \equiv 2 \pmod{3}$ (optimality of $\gamma(m)$); and the orbit-sum identity of Lemma 3.3. Machine-checked *conditionally on Theorem 2.3 and Proposition 2.4*, which are stated as explicit hypotheses: the tower congruences of Lemma 6.1 — including the exact $p = 2$ sign cancellation — and the resulting valuation bounds of Theorem 1.2 in the normalization $n^2 \text{DT}_n = \pm S(n)$, together with the derived p -adic bound at the DT level. Thus the entire reduction of Theorem 1.2 to the classical supercongruence inputs is formally verified; formalizing those inputs themselves (the p -adic Morita gamma arguments of [RZ95] and Appendix B) is left as a further project, the main missing Mathlib ingredients being Volkenborn-integral estimates and Strassmann's theorem.

All code (Python; exact arithmetic throughout) is included with the paper.

APPENDIX A. TABLES

TABLE 2. Numerical DT invariants $\text{DT}_n^{(m)}$, $2 \leq m \leq 6$, $n \leq 10$ (from (3); identical values from both refined implementations at $q = 1$).

$m \backslash n$	1	2	3	4	5	6	7	8	9	10
2	1	1	1	2	5	13	35	100	300	925
3	1	1	3	10	40	171	791	3828	19287	100140
4	1	2	6	28	155	936	6041	41080	290565	2119190
5	1	2	10	60	425	3296	27447	240312	2188056	20544450
6	1	3	15	110	950	9021	91763	982652	10942254	125656950

TABLE 3. The optimal improved-integrality constant $\gamma(m) = 2^{\varepsilon_2(m)} 3^{\varepsilon_3(m)}$ of Theorem 1.2.

m	2	3	4	5	6	7	8	9	10	11	12
$\gamma(m)$	6	2	1	3	2	2	3	1	2	6	1

APPENDIX B. PROOF OF THE SIGNED KAZANDZIDIS CONGRUENCE AT $p = 2$

We prove Proposition 2.4. Throughout, $N \geq K \geq 1$, $M := N - K \geq 1$, $v := v_2$, and $\log$ denotes the 2-adic logarithm, an isometric isomorphism $1 + 8\mathbb{Z}_2 \rightarrow 8\mathbb{Z}_2$ (and defined on $1 + 4\mathbb{Z}_2$ with $v(\log u) = v(u - 1)$ for $u \in 1 + 8\mathbb{Z}_2$).

Step 1: reduction to odd double factorials. Let $G(n) := \prod_{j=1}^n (2j-1) = (2n-1)!!$. From $(2n)! = 2^n n! G(n)$ we get

$$(10) \quad Q := \binom{2N}{2K} / \binom{N}{K} = \frac{G(N)}{G(K)G(M)} \in \mathbb{Z}_2^\times,$$

and $\binom{2N}{2K} - (-1)^{KM} \binom{N}{K} = \binom{N}{K} (Q - (-1)^{KM})$. Since $v\binom{N}{K} \geq 0$, Proposition 2.4 is equivalent to

$$(11) \quad v(Q - \varepsilon) \geq 1 + v(NKM), \quad \varepsilon := (-1)^{KM}.$$

Step 2: the sign. Each factor $2j-1$ is $\equiv 1 \pmod{4}$ if j is odd and $\equiv 3 \pmod{4}$ if j is even; hence $G(n) = (-1)^{\lfloor n/2 \rfloor} H(n)$ with $H(n) := \prod_{j \leq n} ((-1)^{j+1} (2j-1)) \in 1 + 4\mathbb{Z}_2$. Moreover

$$\left\lfloor \frac{N}{2} \right\rfloor - \left\lfloor \frac{K}{2} \right\rfloor - \left\lfloor \frac{M}{2} \right\rfloor \equiv KM \pmod{2},$$

as one checks in the two cases: if K, M are both odd then N is even and the left side is $\frac{N}{2} - \frac{K-1}{2} - \frac{M-1}{2} = 1$; otherwise, say K even, $\lfloor K/2 \rfloor + \lfloor M/2 \rfloor = \lfloor N/2 \rfloor$. Therefore

$$(12) \quad Q = \varepsilon \frac{H(N)}{H(K)H(M)}, \quad \frac{H(N)}{H(K)H(M)} \in 1 + 4\mathbb{Z}_2.$$

In particular $v(Q - \varepsilon) \geq 2$ and $v(Q + \varepsilon) = 1$, so from $Q^2 - 1 = (Q - \varepsilon)(Q + \varepsilon)$,

$$(13) \quad v(Q - \varepsilon) = v(Q^2 - 1) - 1.$$

Step 3: an interpolation lemma. Note that $Q^2 = G(N)^2 / (G(K)^2 G(M)^2)$ and $(2j-1)^2 \in 1 + 8\mathbb{Z}_2$, so with

$$g(y) := \sum_{j=1}^y \log((2j-1)^2) \quad (y \in \mathbb{Z}_{\geq 0})$$

we have, by (13) and the isometry of $\log$ on $1 + 8\mathbb{Z}_2$,

$$(14) \quad v(Q^2 - 1) = v(D), \quad D := g(N) - g(K) - g(M).$$

Lemma B.1. *There is a power series $g(y) = \sum_{i \geq 1} g_i y^i \in \mathbb{Q}_2[[y]]$, convergent on $\mathbb{Z}_2$, which agrees with the sums above at all $y \in \mathbb{Z}_{\geq 0}$, and satisfies:*

- (1) $g_i = 0$ for all even i ;
- (2) $v(g_i) \geq 2$ for all odd $i \geq 3$.

Proof. The function g satisfies $g(0) = 0$ and the difference equation

$$g(y+1) - g(y) = h(y) := \log((1+2y)^2) = \sum_{r \geq 1} h_r y^r, \quad h_r = \frac{2(-1)^{r+1} 2^r}{r},$$

where $v(h_r) = r+1 - v(r) \rightarrow \infty$. Using Faulhaber's formula $\sum_{k=0}^{y-1} k^r = \frac{B_{r+1}(y) - B_{r+1}}{r+1}$ (Bernoulli polynomials, $B_1 = -\frac{1}{2}$) we may define

$$(15) \quad g(y) := \sum_{r \geq 1} \frac{h_r}{r+1} (B_{r+1}(y) - B_{r+1}) = \sum_{i \geq 1} \left(\sum_{r \geq \max(i-1, 1)} \frac{h_r}{r+1} \binom{r+1}{i} B_{r+1-i} \right) y^i.$$

Since $v(h_r) \rightarrow \infty$ and, by von Staudt–Clausen, $v(B_{2s}) \geq -1$ (while $B_1 = -\frac{1}{2}$ and $B_{\text{odd} \geq 3} = 0$), the double sum converges coefficientwise and uniformly on $\mathbb{Z}_2$; termwise summation of the difference equation shows the resulting series interpolates the integer sums.

(1) The interpolated function is odd. First note that the difference equation $g(y+1) - g(y) = h(y)$, verified above at nonnegative integers by construction, holds identically on $\mathbb{Z}_2$: the convergent series $g(y+1) - g(y) - h(y)$ vanishes at all $y \in \mathbb{Z}_{\geq 0}$, hence identically by Strassmann's

theorem (a nonzero convergent power series on $\mathbb{Z}_2$ has finitely many zeros). Iterating it backwards from $g(0) = 0$ gives, for $n \geq 1$, $g(-n) = -\sum_{j=1}^n h(-j) = -\sum_{j=1}^n \log((1-2j)^2) = -g(n)$. Hence the convergent series $g(y) + g(-y)$ vanishes at every nonnegative integer, so again by Strassmann $g(y) + g(-y) \equiv 0$. Thus $g_i = 0$ for even i .

(2) Fix odd $i \geq 3$ and bound each term of the inner sum in (15), using $v(h_r) = r + 1 - v(r)$:

- $r = i - 1$ (so $B_0 = 1$): the term is $\frac{h_{i-1}}{i}$, of valuation $i - v(i-1) - v(i) = i - v(i-1) \geq 2$, since i is odd and $2^{v(i-1)} \leq i - 1$ forces $v(i-1) \leq \log_2(i-1) \leq i - 2$.
- $r = i$ (so $B_1 = -\frac{1}{2}$): valuation $(i+1 - v(i)) + v\binom{i+1}{i} - 1 - v(i+1) = i \geq 3$, using $v(i) = 0$ and $v\binom{i+1}{i} = v(i+1)$.
- $r > i$ with $B_{r+1-i} \neq 0$: then $r+1-i$ is even, so r is even (as i is odd), $r \geq i+1 \geq 4$, and the valuation is at least $(r+1 - v(r)) - 1 - v(r+1) + 0 = r - v(r) \geq r - \log_2 r \geq 2$.

Hence $v(g_i) \geq 2$ for odd $i \geq 3$, proving (2). $\square$

Step 4: assembly. By Lemma B.1 and (14), writing $c_i := g_i$,

$$D = \sum_{i \text{ odd}} c_i(N^i - K^i - M^i) = \sum_{i \text{ odd}, i \geq 3} c_i(N^i - K^i - M^i),$$

the $i = 1$ term vanishing because $N = K + M$. For odd $i \geq 3$, the polynomial $(K + M)^i - K^i - M^i \in \mathbb{Z}[K, M]$ vanishes on each of the hyperplanes $K = 0$, $M = 0$, $K + M = 0$ (the last because i is odd), hence is divisible by $KM(K + M)$ in $\mathbb{Z}[K, M]$; therefore $v(N^i - K^i - M^i) \geq v(NKM)$. Combining with $v(c_i) \geq 2$:

$$v(D) \geq \min_{i \text{ odd}, i \geq 3} (v(c_i) + v(NKM)) \geq 2 + v(NKM).$$

By (13) and (14), $v(Q - \varepsilon) \geq 1 + v(NKM)$, which is (11). $\square$

Remark B.2. The bound is sharp: at $(N, K) = (2, 1)$ one has $Q = 3$, $\varepsilon = -1$ and $v(Q - \varepsilon) = 2 = 1 + v(NKM)$. In the proof this traces to the term $r = 2$, $i = 3$ of (15), whose valuation 2 is exactly attained ($v(g_3) = 2$). We verified Proposition 2.4 numerically for all $2 \leq N \leq 300$, $1 \leq K < N$ (44,850 cases): no violations, and the minimum over all cases of $v\left(\binom{2N}{2K} - \varepsilon \binom{N}{K}\right) - 1 - v(NKM \binom{N}{K})$ equals 0. The coefficient bounds of Lemma B.1 were checked symbolically for $i \leq 10$: $v(g_3) = 2$, $v(g_5) = 3$, $v(g_7) = 6$, $v(g_9) = 6$, and the even coefficients vanish.

REFERENCES

- [AZ06] G. Almkvist, W. Zudilin, *Differential equations, mirror maps and zeta values*, in: Mirror Symmetry V, AMS/IP Stud. Adv. Math. **38** (2006), 481–515.
- [An99] G. E. Andrews, *q-analogs of the binomial coefficient congruences of Babbage, Wolstenholme and Glaisher*, Discrete Math. **204** (1999), 15–25.
- [BCM17] E. Basor, B. Conrey, K. E. Morrison, *Knots and ones*, arXiv:1703.00990.
- [CLPZ23] Q. Chen, K. Liu, P. Peng, S. Zhu, *Congruent skein relations for colored HOMFLY-PT invariants*, Comm. Math. Phys. **400** (2023), 683–729.
- [Cl95] W. E. Clark, *q-analogue of a binomial coefficient congruence*, Internat. J. Math. Math. Sci. **18** (1995), 197–200.
- [Ef12] A. I. Efimov, *Cohomological Hall algebra of a symmetric quiver*, Compos. Math. **148** (2012), 1133–1146. arXiv:1103.2736.
- [GKS16] S. Garoufalidis, P. Kucharski, P. Sułkowski, *Knots, BPS states, and algebraic curves*, Comm. Math. Phys. **346** (2016), 75–113. arXiv:1504.06327.
- [GSWZ24] S. Garoufalidis, P. Scholze, C. Wheeler, D. Zagier, *The Habiro ring of a number field*, arXiv:2412.04241.
- [Go19] O. Gorodetsky, *q-congruences, with applications to supercongruences and the cyclic sieving phenomenon*, Int. J. Number Theory **15** (2019), 1919–1968. arXiv:1805.01254.
- [Gos24] F. Gossow, *Polynomial and combinatorial analogues of Gauss congruence*, arXiv:2410.05678.

- [Kaz68] G. S. Kazandzidis, *Congruences on the binomial coefficients*, Bull. Soc. Math. Grèce (N.S.) **9** (1968), 1–12; **10** (1969), 35–40.
- [KS11] M. Kontsevich, Y. Soibelman, *Cohomological Hall algebra, exponential Hodge structures and motivic Donaldson–Thomas invariants*, Commun. Number Theory Phys. **5** (2011), 231–352. arXiv:1006.2706.
- [KSV06] M. Kontsevich, A. Schwarz, V. Vologodsky, *Integrality of instanton numbers and p -adic B-model*, Phys. Lett. B **637** (2006), 97–101. hep-th/0603106.
- [KRSS19] P. Kucharski, M. Reineke, M. Stošić, P. Sułkowski, *Knots-quivers correspondence*, Adv. Theor. Math. Phys. **23** (2019), 1849–1902. arXiv:1707.04017.
- [KS16] P. Kucharski, P. Sułkowski, *BPS counting for knots and combinatorics on words*, JHEP **11** (2016) 120. arXiv:1608.06600.
- [LZ16] W. Luo, S. Zhu, *Integrality structures in topological strings I: framed unknot*, arXiv:1611.06506.
- [MR19] S. Meinhardt, M. Reineke, *Donaldson–Thomas invariants versus intersection cohomology of quiver moduli*, J. reine angew. Math. **754** (2019), 143–178. arXiv:1411.4062.
- [Me11] R. Meštrović, *Wolstenholme’s theorem: its generalizations and extensions in the last hundred and fifty years (1862–2012)*, arXiv:1111.3057.
- [Mü21] L. F. Müller, *Wolstenholme type congruences and framing of rational 2-functions*, arXiv:2104.10754.
- [Pan13] H. Pan, *Factors of some lacunary q -binomial sums*, Monatsh. Math. **172** (2013), 387–398.
- [PSS18] M. Panfil, M. Stošić, P. Sułkowski, *Donaldson–Thomas invariants, torus knots, and lattice paths*, Phys. Rev. D **98** (2018), 026022. arXiv:1802.04573.
- [Re05] M. Reineke, *Cohomology of non-commutative Hilbert schemes*, Algebr. Represent. Theory **8** (2005), 541–561. arXiv:math/0306185.
- [Re11] M. Reineke, *Cohomology of quiver moduli, functional equations, and integrality of Donaldson–Thomas type invariants*, Compos. Math. **147** (2011), 943–964. arXiv:0903.0261.
- [Re12] M. Reineke, *Degenerate cohomological Hall algebra and quantized Donaldson–Thomas invariants for m -loop quivers*, Doc. Math. **17** (2012), 1–22. arXiv:1102.3978.
- [Re24] M. Reineke, *Donaldson–Thomas invariants of symmetric quivers*, survey, arXiv:2410.03219.
- [RZ95] A. Robert, M. Zuber, *The Kazandzidis supercongruences. A simple proof and an application*, Rend. Sem. Mat. Univ. Padova **94** (1995), 235–243.
- [Ro00] A. Robert, *A Course in p -adic Analysis*, Grad. Texts in Math. **198**, Springer, 2000.
- [SVW17] A. Schwarz, V. Vologodsky, J. Walcher, *Integrality of framing and geometric origin of 2-functions*, arXiv:1702.07135.
- [St11] A. Straub, *A q -analog of Ljunggren’s binomial congruence*, DMTCS Proc. FPSAC 2011, 897–902. arXiv:1103.3258.
- [St19] A. Straub, *Supercongruences for polynomial analogs of the Apéry numbers*, Proc. Amer. Math. Soc. **147** (2019), 1023–1036. arXiv:1803.07146.
- [WZ25] W. Wang, S. Zhu, *BPS invariants from framed links*, arXiv:2502.16609.
- [Zu19] W. Zudilin, *Congruences for q -binomial coefficients*, Ann. Comb. **23** (2019), 1123–1135. arXiv:1901.07843.